

Perfect substitutes

Consumer theory · Problem 1 · Video: youtu.be/qKRRKmEKNqFQ

Solved problem

A student has a weekly food budget of $M = \$20$ to spend on two goods:

- x : packets of instant noodles, at $p_x = \$4$ each;
- y : cups of fruit yogurt, at $p_y = \$5$ each.

Preferences are represented by $u(x, y) = x + 2y$, so one more cup of yogurt is worth twice as much as one more packet of noodles. Find the optimal bundle (x^*, y^*) and illustrate the budget line and the optimal choice in a diagram.

Solution

Here we solve the following problem:

$$\begin{aligned} \max_{(x,y) \in \mathbb{R}_+^2} \quad & x + 2y \\ \text{s.t.} \quad & 4x + 5y \leq 20 \end{aligned}$$

Compare the marginal utility per dollar spent on each good:

$$\frac{MU_x}{p_x} = \frac{1}{4} \quad \text{and} \quad \frac{MU_y}{p_y} = \frac{2}{5}.$$

Since $MU_x/p_x < MU_y/p_y$, every dollar buys more utility on yogurt, so the student spends the whole budget on y . With $M = 20$ and $p_y = 5$, the optimal bundle is

$$(x^*, y^*) = \left(0, \frac{M}{p_y}\right) = (0, 4).$$

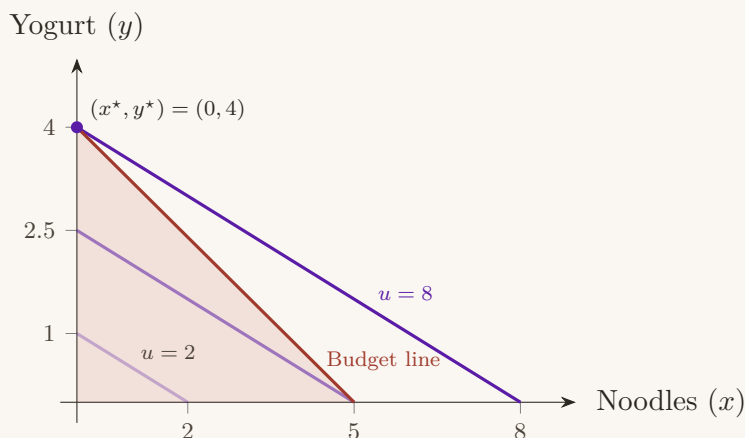


Figure 1. Indifference curves are straight lines $x + 2y = \bar{u}$ with absolute slope $MU_x/MU_y = 1/2$. The budget line has absolute slope $p_x/p_y = 4/5$ and is steeper, so the highest attainable indifference curve is reached at the corner $(0, 4)$, where the budget line meets the y -axis.

Exercise

A student commutes to university by either:

- x : bus rides;
- y : metro rides.

Each ride, by either mode, is one trip to university, so the student treats the two modes as perfect substitutes:

$$u(x, y) = x + y.$$

A bus ride costs $p_x = \$20$ and a metro ride $p_y = \$30$, and the weekly transport budget is $M = \$180$.

1. Draw the budget set and the budget line.
2. Compute the marginal utility per dollar spent on each mode, $\frac{MU_x}{p_x}$ and $\frac{MU_y}{p_y}$. Find the utility-maximising bundle (x^*, y^*) .
3. Suppose the metro authority introduces a discounted student fare, cutting the metro price to $p_y = \$20$. With everything else unchanged, characterise the set of utility-maximising bundles.

Extension: quality-adjusted travel

Suppose instead the student finds the metro more convenient, because it is faster and more reliable, so preferences become

$$u(x, y) = x + 1.5y,$$

with the original prices and budget, $p_x = \$20$, $p_y = \$30$, and $M = \$180$.

4. Solve for the utility-maximising bundle(s) under these preferences.

Checking your work. To discuss the exercise, join the community forum at econschool.in.